

0. Group Members

Hardik Shrestha – 251163014

Solo

1. Project Title

Dissecting King of the Hill Strategy: A Comparative Study of Neural Network vs. Heuristic Evaluation in the Early-Middlegame

2. Dataset Link

Preferred: [lichess.org open database](https://lichess.org/open-database) (ctrl+f king of the hill)

Backup: [Lichess/king-of-the-hill-chess-games · Datasets at Hugging Face](https://huggingface.co/datasets/Lichess/king-of-the-hill-chess-games)

3. Project Description

The **King of the Hill (KotH)** chess variant is characterized by complex positional sacrifices and King maneuvers, particularly during the mid-game. This project will conduct a highly focused comparative study of evaluation techniques, specifically targeting board positions between **moves 10 and 20** (the early-middlegame). We will implement two distinct evaluation functions: 1) a traditional, **hand-tuned heuristic** (Minimax/Alpha-Beta based) that penalizes King exposure, and 2) a specialized **NNUE (Efficiently Updatable Neural Network)** trained on millions of KotH games from the Lichess database. The core engineering problem is determining which evaluation method better captures the unique, central-King dynamics of the KotH middlegame. The study will involve using the trained NNUE to re-evaluate key middlegame positions and compare its calculated winning probability against the heuristic's centipawn score, specifically analyzing positions where the two models **disagree** by more than a set threshold (estimating ~200 cp). This narrow focus ensures **extensiveness of analysis** by dedicating all resources to interpreting and explaining the strategic biases of the learned model.